

Ahsanullah University of Science and Technology



Department of Electrical and Electronics Engineering

OEL Report

Course no : EEE 3104

Course name: Digital Electronics-1 Lab

**Project name: Digital Voting System Using Sequential Logic
Circuits**

Submitted By:

Group:04

Section:B1

Student ID:

20220105082

20220105083

20220105084

20220105085

20220105089

Introduction:

The digital voting system presented in this experiment utilizes sequential logic circuits, primarily consisting of flip-flops, counters, comparators, and seven-segment displays to count and display votes. This system is designed to register votes from multiple candidates, increment their respective counts, and display results in real-time. The circuit ensures reliability, accuracy, and security in the voting process.

Objectives

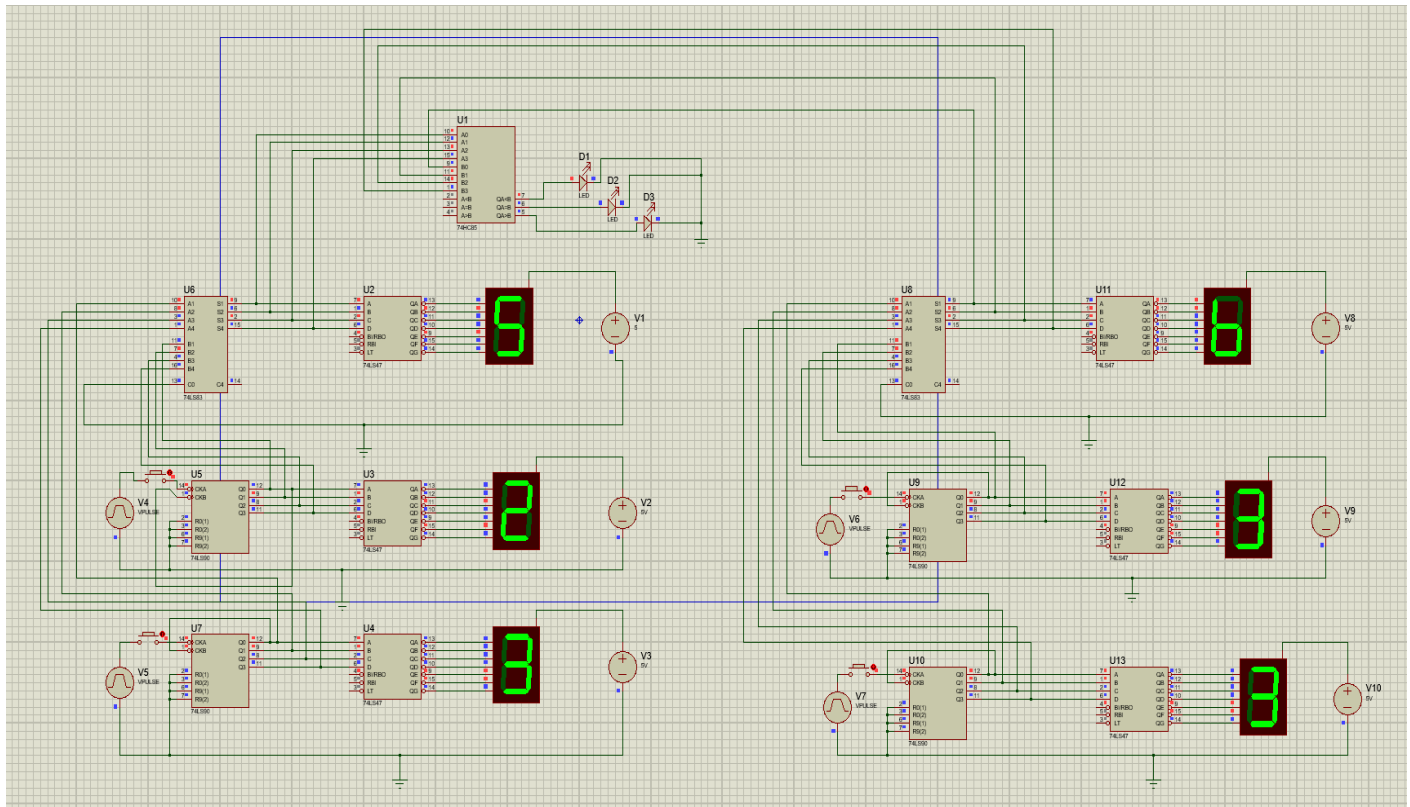
- To design a digital voting system using logic circuits.
- To implement a counting mechanism for each candidate.
- To display vote counts on seven-segment displays.
- To ensure accurate vote registration using debounced switches.

Equipments:

- 74LS90 (Decade Counter)
- 74LS47 (BCD to Seven-Segment Decoder)
- 74LS83 (4-bit Binary Adder)
- 74HC85 (4-bit Comparator)
- Seven-Segment Displays
- Switches (for vote input)

- LEDs (status indicators)
- Resistors and Capacitors
- Power Supply (5V DC)

Circuit Diagram:



Working Principle:

The voting system operates based on the following key components and their interactions:

- **Input Section:** Each voter registers their vote using a momentary push-button switch assigned to a specific candidate. The switches are debounced to prevent erroneous multiple triggers.
- **Counter Mechanism:** Upon pressing a vote switch, a clock pulse is sent to the corresponding 74LS90 decade counter. This counter increments the vote count for that candidate

and outputs the count in Binary-Coded Decimal (BCD) format.

- **Decoding and Display Section:** The BCD output from each counter is fed into a 74LS47 BCD-to-seven-segment decoder, which converts the binary input into a decimal display on a seven-segment LED. This provides a clear real-time tally of votes.
- **Comparator Logic:** A 74HC85 4-bit comparator is used to compare vote counts between candidates. This allows for real-time identification of the candidate with the highest votes, which could be indicated using LEDs.

Observations :

During testing, the system accurately recorded votes, displayed correct values on seven-segment displays, and effectively debounced input switches to avoid multiple false counts. The reset function successfully cleared all counters. The comparator functioned correctly, identifying the highest vote count in real time.

Advantages and Limitations Advantages:

- Real-time vote counting
- Simple and reliable circuit design
- Debounced switches to prevent errors
- Easily expandable for more candidates
- Real-time vote comparison using 74HC85

Discussion:

The digital voting system developed in this experiment provides a basic framework for secure and reliable vote counting. While the design effectively registers and displays votes, there are several areas for improvement. One key consideration is the lack of voter authentication, which makes the system vulnerable to multiple voting by a single person. Additionally, while the use of the 74HC85 comparator enables real-time vote comparison, more advanced methods such as microcontroller-based processing could allow for greater flexibility and security.

Another factor to consider is scalability. The current design is limited by the number of available counters and displays, making it less practical for large-scale elections. Future implementations could incorporate a microcontroller or FPGA to handle multiple candidates more efficiently. Moreover, integrating EEPROM or cloud storage would allow for vote retention after power loss, ensuring data integrity.

Overall, while the system is effective for small-scale voting applications, improvements in security, scalability, and data retention could enhance its practical usability for more complex voting scenarios.

Conclusion:

The designed digital voting system successfully registers, counts, and displays votes using sequential logic circuits. This system provides an efficient and error-free method for conducting small-scale elections or polls. Future improvements could include microcontroller-based implementations for authentication and remote voting capabilities.